

High Package (LPA) Hardware + Firmware + Linux + AI + Robotics + Communication Protocols
+ Real-time Systems strong grip .

Project Roadmap follow strong portfolio build

Beginner Level Projects (Foundation Strong)

1. Smart Home Automation

- ESP32
- Relay
- Mobile App Control
- WiFi + Bluetooth

Skills:

- GPIO
 - Sensors
 - MQTT
 - Embedded C
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2. Line Following Robot

- IR Sensors
- Motor Driver
- PID Control

Skills:

- Robotics basics
 - Motor control
 - Sensor calibration
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3. Smart Irrigation System

- Soil Moisture Sensor
- Pump Automation
- IoT Dashboard

Skills:

- IoT
 - Sensor interfacing
 - Cloud basics
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Intermediate Level Projects

4. RTOS Based Industrial Monitoring System

Use:

- FreeRTOS
- ESP32 / STM32

Features:

- Multitasking
- Sensor reading
- Task scheduling
- Real-time alerts

Skills:

- RTOS
 - Scheduling
 - Interrupts
 - Queues/Semaphores
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5. Raspberry Pi CCTV AI Camera

Features:

- Live streaming
- Motion detection
- Human detection
- Night vision

Skills:

- Linux
 - OpenCV
 - Python
 - Computer Vision
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6. Vehicle Tracking System

- GPS + GSM
- Real-time location
- Geo fencing

Skills:

- UART
 - GPS protocol
 - GSM communication
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7. Custom PCB Design Project

Tools:

- KiCad
- EasyEDA

Skills:

- PCB layout
- Circuit design
- Power management

Very important for high salary roles.

Advanced Level Projects (High Package Projects)

8. Drone Flight Controller

Features:

- Gyroscope stabilization
- PID balancing
- GPS navigation

Skills:

- STM32
- IMU sensors
- Real-time control systems

Very valuable project.

9. Autonomous Rover Robot

Features:

- Obstacle avoidance
- Camera navigation
- AI object detection

Skills:

- ROS
 - SLAM
 - Computer Vision
 - Sensor fusion
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10. Embedded Linux Development Board

Build:

- Custom Linux image
- Device drivers
- Bootloader

Skills:

- Yocto
- Linux kernel
- Drivers
- UART/SPI/I2C

This is what top companies want.

11. AI + Embedded Edge Device

Example:

- Face recognition door lock
- Thermal human detector
- Smart surveillance

Hardware:

- Jetson Nano
- Raspberry Pi AI Kit

Skills:

- TinyML
 - TensorFlow Lite
 - Edge AI
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12. Automotive ECU Simulation

Features:

- CAN Communication
- Diagnostics
- Sensor emulation

Skills:

- CAN protocol
- Automotive embedded
- AUTOSAR basics

Very high salary domain.

Most Important Skills

Programming

- C (mandatory)
 - Embedded C
 - C++
 - Python
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Protocols

- UART
 - SPI
 - I2C
 - CAN
 - MQTT
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Hardware Knowledge

- PCB Design
 - Power Electronics basics
 - Oscilloscope usage
 - Multimeter debugging
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Software Tools

- STM32CubeIDE

- VS Code
 - PlatformIO
 - KiCad
 - MATLAB basics
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High Salary Domains in Embedded

Domain	Salary Growth
Automotive Embedded	Very High
Robotics	Very High
Drone Systems	High
Medical Embedded	High
Industrial IoT	High
Embedded AI	Extremely High

Best Hardware To Buy

Beginner

- ESP32
- Arduino UNO

Intermediate

- Raspberry Pi
- STM32

Advanced

- Jetson Nano
- NVIDIA Orin Nano

Companies That Pay High Packages

- Qualcomm
 - NVIDIA
 - Texas Instruments
 - Bosch
 - Intel
 - Samsung
 - AMD
 - NXP
 - Tata Elxsi
 - Continental
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Ultimate Roadmap

Step 1

Learn:

- C Programming
- Electronics basics

Step 2

Build:

- ESP32 Projects

Step 3

Learn:

- RTOS
- Communication protocols

Step 4

Build:

- Robotics + AI projects

Step 5

Learn:

- Embedded Linux
- Device drivers

Step 6

Create:

- Strong GitHub portfolio
- LinkedIn project showcase